Monojit Layek

Robotics Motion Planning Engineer

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Summary

Robotics and Automation M.Tech graduate from IIT Kanpur with expertise in C++ motion-planning stacks, kinematic modeling, and trajectory optimization. Experienced in developing collision-aware planners (RRT/RRT*), inverse kinematics solvers, and velocity profiles using ROS 2, MoveIt 2, Gazebo, and PyBullet. Strong mathematical foundation in constrained nonlinear optimization and 6-DOF robot dynamics for reliable real-time autonomy.

Experience

6–DOF STEWART PLATFORM

AeThrone Aerospace | System Design Scientiest

Mar 2023 – June 2024 | Pune, India

- Designed a **6-DOF Stewart platform** for a ship-mounted drone landing system to compensate for vessel motion and maintain landing-platform stability against **pitch, roll, and yaw** disturbances.
- Developed the platform's **CAD model and URDF description** for robotic modeling and simulation.
- Developed robotic software incorporating **forward and inverse kinematics** and dynamic analysis, using **ROS** for framework integration and **PyBullet** for simulation-based evaluation of platform motion and stabilization behavior.

6-AXIS DOBOT MANIPULATOR – MOTION PLANNING STACK

Personal Project | C++ • ROS 2 • MoveIt 2

Oct 2022 | UP, India

- Built a full **ROS 2 Humble** motion-planning stack in **C++** for a 6-axis manipulator, featuring a **Damped Least Squares IK solver** with geometric Jacobian and singularity-robust pseudo-inverse.
- Designed a **trajectory planning engine** with **Cubic Spline** interpolation and **Trapezoidal / Quintic-Polynomial** velocity profiling for smooth, jerk-free motion.
- Integrated a **collision-aware planner** (RRT/RRT*, MoveIt 2) for a Pick-and-Place pipeline; validated in **Gazebo** and **PyBullet** simulations.

MECHATRONIC DESIGN & CLOSED-LOOP CONTROL OF A MAGNETIC RECIPROCATING ACTUATOR [MRE]

Mechanical Design and Research Project | KGEC

Jul 2019 | WB, India

- Modeled the mechanism as **a constrained slider-crank system**, performing **analytical dynamic torque profiling** and **simulating the kinematic conversion** of linear reciprocation to continuous rotary output.
- Implemented a **sensor-driven closed-loop control system** on an Arduino micro-controller, integrating **photoelectric position feedback** and solid-state switching to automate **phase-locked electromagnetic actuation**.
- Designed the full **CAD architecture** and built the **physical hardware prototype** to benchmark empirical motion stability and torque output against theoretical models.

Skills

PROGRAMMING

• C++ • Python • MATLAB

ROBOTICS FRAMEWORKS & SIMULATION

• ROS 2 • MoveIt 2 • PyBullet • Gazebo • URDF

3D MODELING & DEVELOPMENT TOOLS

• AutoCad • SolidWorks • Fusion • Git • Linux

Education

IIT KANPUR

MTech in Robotics & Automation

Jun 2023 | Kanpur, UP, India

KALYANI GOVERNMENT ENGINEERING COLLEGE

BTech in Mechanical Engineering

Jun 2019 | Kalyani, WB, India

Graduated with Distinction

Coursework

POSTGRADUATE

• Introduction to Robotics
• Motion Planning
• Numerical Optimization
• Computer-Aided Design (CAD)
• Advanced Mathematics
• Deep Learning for Computer Vision

Dissertation: Kanpur Version of Zoutendijk Method for Constrained Nonlinear Optimization (Co-authored with IIT Kanpur & IISc Bangalore; benchmarked up to 13-DOF constrained spaces).

Certification

1. Deep Learning Specialization
   • Feb 2023 | Deeplearning-ai
2. ML Specialization
   • Aug 2021 | Stanford
3. Data-Structure Algorithm
   • Jul 2021 | Udemy

KANPUR-ZOUTENDIJK CONSTRAINED NONLINEAR OPTIMIZATION SOLVER

Research Publication | Indian Institute of Technology (IIT) Kanpur

June 2024 | UP, India

- Co-authored a **feasible-direction nonlinear optimization solver** that projects the Simplex LP subproblem from the n-dimensional state space onto the m-dimensional **active-constraint subspace ($m \leq n$)**, reducing computational complexity from $\mathcal{O}(N^2)$ to $\mathcal{O}(N\log(N))$ and accelerating **CPU runtimes by 2–3.1x**.
- Designed a **Newton-step infeasible-start recovery routine** and a **shrinking trust-region strategy** for equality constraints, outperforming classic Zoutendijk and evolutionary baselines across **2D–13D non-convex benchmark spaces** with tighter **KKT convergence (E_{KKT})**.